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School of *Computer Science & Mathematics*

BSc DEGREE IN *Computer Science* PROJECT PROPOSAL

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Did you discuss and agree the viability of your project idea with your supervisor?	Yes
Did you submit a draft of your proposal to your supervisor?	No
Did you receive feedback from your supervisor on any submitted draft?	No

Abstract

The chilli industry faces challenges due to reliance on manual processes, supply chain inefficiencies, lack of real-time pricing data, and limited transparency. This project addresses these issues by developing a Chilli Management System, a digital platform designed to optimize chilli production, supply, and pricing. The system integrates user authentication and role-based dashboards for administrators, suppliers, and buyers, enabling efficient management and access to relevant information.

A key component of the system is a price prediction system that utilizes machine learning algorithms to forecast chilli prices based on historical data and market trends. This feature aims to provide price visibility and support fair trade decisions. The system also includes inventory management functionality to track stock levels and a supplier-buyer communication module to facilitate direct interaction and negotiations.

The system is designed using a multi-tiered architecture, with a focus on user-friendly interfaces and robust data management. The implementation involves technologies such as [I'll need to specify technologies used, e.g., Python, a web framework, and a database]. The system's performance and usability will be evaluated through testing and user feedback.

The Chilli Management System is expected to enhance market efficiency, improve price stability and forecasting, increase transparency, and reduce transaction costs. By digitizing the chilli market and providing tools for better decision-making, the project contributes to the development of a more efficient, transparent, and profitable chilli supply chain.

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Glossary of Terms

- GDP - Gross Domestic Product
- ARIMA - Autoregressive Integrated Moving Average
- ANNs - Artificial Neural Networks
- ML - Machine learning
- CSS - Cascading Style Sheets
- UI - User Interface
- SRS - Software Requirements Specification
- UAT - User Acceptance Testing

1. Introduction & Background

1.1 Introduction

The agricultural sector underpins many economies worldwide-and in Sri Lanka, it not only employs a large share of the workforce but also makes a significant contribution to GDP. Despite its importance, farmers and consumers alike face substantial hardship due to volatile crop prices. In the chilli market especially, manual inventory controls, opaque supply chains, and the lack of real-time pricing data compound these challenges, often resulting in economic instability and inefficient decision-making.

To help stabilize prices and streamline operations, this project proposes a modern, data-driven Chilli Management System. By leveraging advanced data-mining techniques and machine-learning algorithms-proven in numerous studies to outperform traditional time-series models like ARIMA-the system will deliver accurate price forecasts, automated inventory tracking, and seamless transaction management. Core functionalities will include role-based access for administrators, suppliers, and buyers; a neural-network-based price prediction engine; an intuitive inventory dashboard; and an integrated communication module to foster transparent, direct interactions.

1.2 Background and Motivation

The agricultural sector is a cornerstone of many economies, including that of Sri Lanka, where it employs a significant portion of the labor force and contributes substantially to the national GDP. However, this sector faces significant challenges, particularly substantial price fluctuations in agricultural products. These fluctuations can distress farmers and consumers and lead to economic instability.

Traditional methods in the agricultural market, especially in the chilli sector, rely heavily on manual processes. This can lead to complications in inventory tracking, price prediction, and transaction management. Inefficient supply chain management, the absence of real-time pricing data, and limited transparency between suppliers and buyers further exacerbate these issues.

To address these challenges, there's a growing need for advanced technological solutions, specifically the application of data mining techniques and machine learning algorithms. These technologies offer the potential to analyze complex agricultural data, predict price variations, and optimize supply chain processes.

Various studies have demonstrated the effectiveness of machine learning in agricultural price prediction. Artificial Neural Networks (ANNs), for instance, have been successfully applied to forecast the prices of various crops, often outperforming traditional time-series models like ARIMA. These models can capture non-linear relationships and provide more accurate predictions, which are crucial for effective decision-making by farmers, suppliers, and other stakeholders.

This project is motivated by the need to modernize the chilli supply chain by developing a chilli management system. The system will integrate key functionalities, including:

- User authentication and role-based dashboards for different users (admin, supplier, and buyer).
- A price prediction system using machine learning algorithms.
- Inventory management to track stock levels.

- A communication module to facilitate direct interaction between suppliers and buyers.

By addressing the issues of price fluctuation, inefficient supply chains, and lack of transparency, the project aims to enhance market efficiency, improve price stability, reduce transaction costs, and support data-driven decision-making in the chilli industry.

1.3. Problem in brief

The core issue is the inefficiency and lack of optimization in the chilli supply chain, specifically concerning production, supply, and pricing. This inefficiency arises from several factors:

- **Reliance on manual processes:** Traditional methods in the chilli market depend heavily on manual operations. This makes it hard to keep track of inventory, predict price changes, and manage transactions effectively.
- **Supply chain inefficiencies:** The supply chain lacks smooth operation, leading to problems in getting chilli from producers to consumers.
- **Lack of real-time pricing data:** There's a shortage of up-to-date information on how much chilli is worth at any given time. This makes it difficult for sellers and buyers to agree on fair prices.
- **Limited transparency:** There's a lack of openness in the dealings between those who provide the chilli and those who buy it. This can create distrust and make it harder to build strong business relationships.

In short, the absence of a good management system means the chilli business isn't working as well as it could. This project aims to fix that by creating a system that makes things more organized, predictable, and fair for everyone involved.

2. Aim & Objectives.

2.1 Aim

- The project aims to digitally transform the chilli supply chain by creating efficient and transparent processes for suppliers, buyers, and administrators

2.2 Objectives

- Develop a secure and user-friendly digital Platform for Admins, Suppliers, and Buyers.
- Apply machine learning algorithms to forecast chilli prices based on market trends and historical data.
- Develop price visibility to help buyers and suppliers make fair trade decisions.
- Facilitate direct communication between suppliers and buyers for better negotiations.
- Monitoring system performance and user activity for continuous improvements

3. Technologies & Resources

3.1 Technologies Used

The Chilli Management System will be built on a modern, scalable web and machine-learning stack that balances developer productivity with performance and maintainability:

- **Backend Framework:**
 - **Python 3.9+** – primary language for business logic, data processing and ML orchestration.
 - **FastAPI** – lightweight, high-performance REST API framework with built-in data validation (Pydantic) and automatic OpenAPI documentation.

- **Frontend Framework:**
 - **React 18** – for building a responsive, component-based single-page application.
 - **Tailwind CSS** – utility-first styling for rapid UI development.
- **Machine Learning & Data Science:**
 - **TensorFlow 2.x** – training and serving neural-network-based price-prediction models.
 - **scikit-learn** – auxiliary algorithms (data preprocessing, baseline regressors).
 - **Pandas, NumPy** – data manipulation and numerical computing.
- **Data Storage & Caching:**
 - **PostgreSQL** – relational database for user profiles, inventory and transaction records.

3.2 Resource Requirement

Resource Type	Description
Human Resources	• Supervisor – domain guidance and sign-off
Hardware	• Developer Workstations (Windows) • Cloud VM Instances: – 2 × general-purpose (t3.medium or n1-standard-1) for API and frontend – 1 × GPU-enabled (g4dn.xlarge or n1-standard-4 + T4) for model training • Backup Storage (S3/RDS snapshot)
Software & Licenses	• PyCharm / VS Code (IDE) • PostgreSQL license (open source) • TensorFlow (open source) • Cloud subscription (AWS/GCP credits)
Data Sources	• Historical chili-price datasets (Ministry of Agriculture CSVs, Kaggle) • Market-survey API or web-scraped price feeds (daily updates) • User-generated data (inventory levels, transaction logs)
Other	• SSL Certificate (Let's Encrypt) • Domain name and DNS management

4. Methodology & Work plan

Phase	Activities	Timeline	Deliverables
1. Initiation & Requirements	- Stakeholder interviews - Finalize functional requirements - Set up repos	May 1 – May 15, 2025	- Requirements Specification (SRS) - Project plan
2. Architecture & Design	- System and data-model design - UX/UI wireframes - API contract definitions	May 16 – Jun 5, 2025	- Architecture diagrams - UI mockups
3. Core Backend & Database	- FastAPI skeleton - PostgreSQL schema	Jun 6 – Jun 30, 2025	Backend codebase
4. Frontend Development	- React app setup - Dashboard layouts - Integration with Auth API	Jul 1 – Jul 21, 2025	- Frontend codebase - Demo UI
5. ML Model Development	- Data cleaning & feature engineering - TensorFlow model training - Hyperparameter tuning	Jul 22 – Aug 31, 2025	- Trained model artifacts - Evaluation report
6. Integration & End-to-End Testing	- Integrate price-prediction endpoint - Inventory and messaging modules	Sep 1 – Sep 21, 2025	- Test reports - Tested MVP
7. Deployment & User Acceptance	- Deploy to staging/production - Conduct UAT with suppliers & buyers	Sep 22 – Oct 12, 2025	- Live system - UAT feedback logged
8. Evaluation & Closure	- Collect feedback - Final report & presentation	Oct 13 – Oct 20, 2025	- Final project report - Supervisor sign-off

5. Proposed Solution: Real-Time Decision Support System (RTDSS) for MOM Systems

To address the problems identified, the proposed solution is the development of a Chilli Management System. This system will be a digital platform with the following key features:

- **User Authentication and Role-Based Dashboards:** The system will provide secure access for different users, including administrators, suppliers, and buyers. Each user will have a personalized dashboard with relevant information and functionalities.
- **Price Prediction System:** Machine learning algorithms will be implemented to analyze historical price data and market trends to forecast future chilli prices. This will help suppliers and buyers make informed decisions.
- **Inventory Management:** The system will track chilli stock levels in real-time, enabling efficient management of supply and demand and preventing shortages or oversupply.
- **Supplier-Buyer Communication Module:** A direct communication platform will facilitate seamless interaction, negotiation, and transactions between suppliers and buyers.

The expected outcomes of this system include:

- Enhanced market efficiency.
- Improved price stability and forecasting.
- Increased transparency in transactions.
- Time and cost savings for stakeholders.
- Data-driven decision-making capabilities.
- Streamlined order management and chilli sourcing.
- Strengthened supplier-buyer relationships.

5.1 Suggested Starting Point

To initiate the development of the Chilli Management System, the following actions will be prioritized:

- i. **Detailed Requirements Analysis:** Conduct a thorough analysis of the specific needs and requirements of each user group (administrators, suppliers, and buyers). This involves gathering information on their current processes, pain points, and expectations from the system.
- ii. **Data Collection and Preparation:** Identify and gather relevant historical data on chilli prices, supply, demand, and other market factors. Ensure the data is accurate, complete, and in a suitable format for analysis and machine learning model training.
- iii. **Technology Selection:** Evaluate and select appropriate technologies and tools for system development, including programming languages, databases, machine learning libraries, and web frameworks.
- iv. **Prototype Design:** Develop a preliminary prototype of the system's user interface and key functionalities. This will serve as a basis for user feedback and further refinement of the system design.

6. Discussion

Legal Issues

- **Data Ownership and Privacy:** The system will collect and store data on suppliers, buyers, transactions, and pricing. It's crucial to establish clear legal frameworks regarding data ownership, usage rights, and privacy. Compliance with data protection regulations is essential.
- **Contractual Agreements:** The system may facilitate contracts between suppliers and buyers. Legal implications related to contract formation, enforcement, and dispute resolution should be considered.
- **Regulatory Compliance:** The chilli trade may be subject to specific agricultural and trade regulations. The system should be designed to support compliance with these regulations.

Ethical Issues

- **Fairness and Transparency:** The price prediction system should be designed to promote fair pricing and prevent market manipulation. Algorithms used for price prediction should be transparent and explainable to avoid bias.
- **Data Security and Confidentiality:** Protecting sensitive business data and user information is paramount. Ethical considerations dictate implementing robust security measures to prevent unauthorized access and data breaches.
- **Accessibility and Inclusivity:** The system should be accessible to all stakeholders, regardless of their technical skills or resources. Efforts should be made to bridge the digital divide and ensure inclusivity.

Societal Issues

- **Impact on Smallholder Farmers:** The system should be designed to empower smallholder farmers and improve their access to markets. It should not inadvertently marginalize them or create new forms of dependency.
- **Food Security and Affordability:** Efficient chilli trading can contribute to food security by ensuring a stable supply and reasonable prices. The system's impact on food affordability for consumers should be considered.
- **Employment and Livelihoods:** The system may affect employment patterns in the chilli industry. It's important to analyze potential job displacement and create opportunities for skill development and alternative livelihoods.

Security Issues

- **Cybersecurity Threats:** The system is vulnerable to cybersecurity threats such as hacking, data breaches, and denial-of-service attacks. Robust security measures, including firewalls, intrusion detection systems, and encryption, are necessary.
- **Data Integrity:** Ensuring the accuracy and reliability of data within the system is crucial. Measures should be implemented to prevent data manipulation, errors, and inconsistencies.
- **System Reliability:** The system should be designed to be reliable and resilient to failures. Redundancy and backup mechanisms should be in place to minimize downtime and data loss.

By carefully addressing these legal, ethical, societal, and security issues, the Chilli Management System can be developed and deployed responsibly, maximizing its benefits while mitigating potential risks.

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